

UTX Inc. Internship Portfolio

Mechanical Engineering Intern
Summer 2026
Spencer Levine

Design • Manufacturing • Automation • Testing

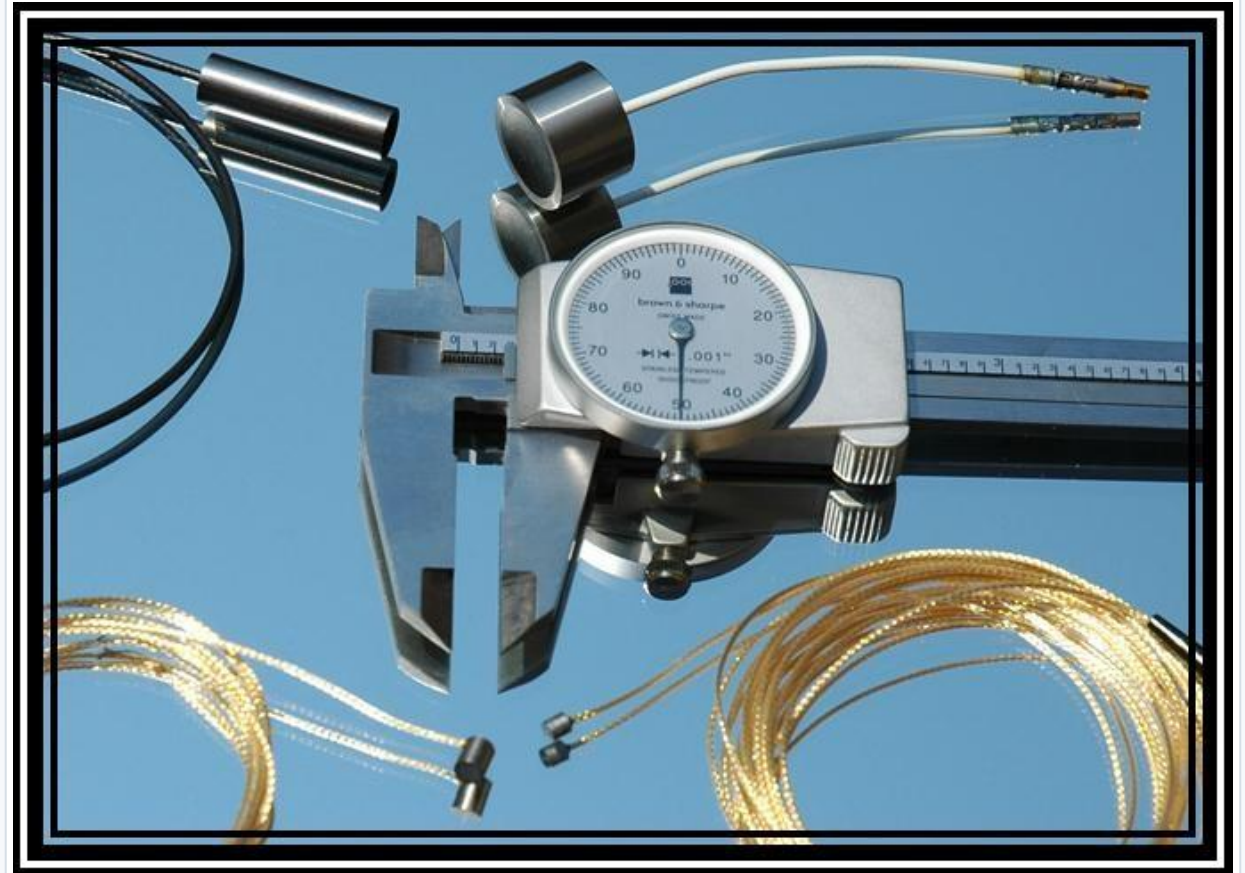


UTX Inc. team • Holmes, New York

What UTX Builds

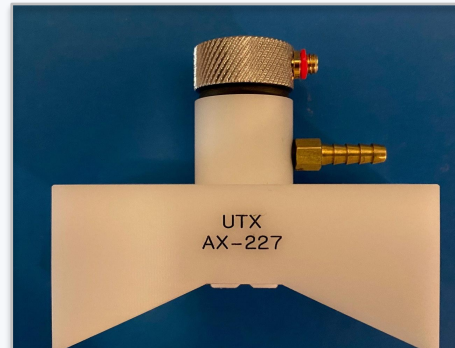
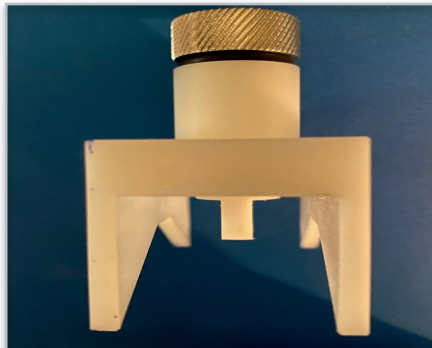
A quick look at the product I spent the summer designing around, manufacturing, and testing.

- UTX Inc. designs and manufactures ultrasonic transducers.
- These transducers use sound waves for inspection, measurement, and testing of things like airplanes, nuclear reactors, and bridges.
- My internship exposed me to both the design and manufacturing sides of the product.

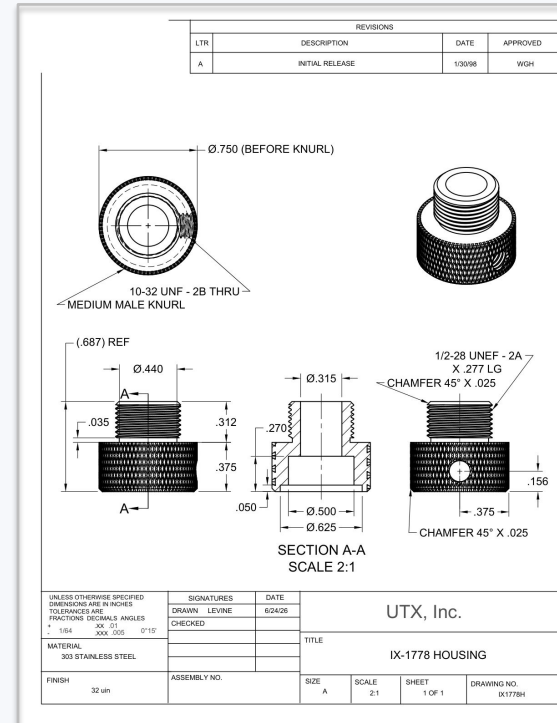


Updating Legacy Engineering Drawings

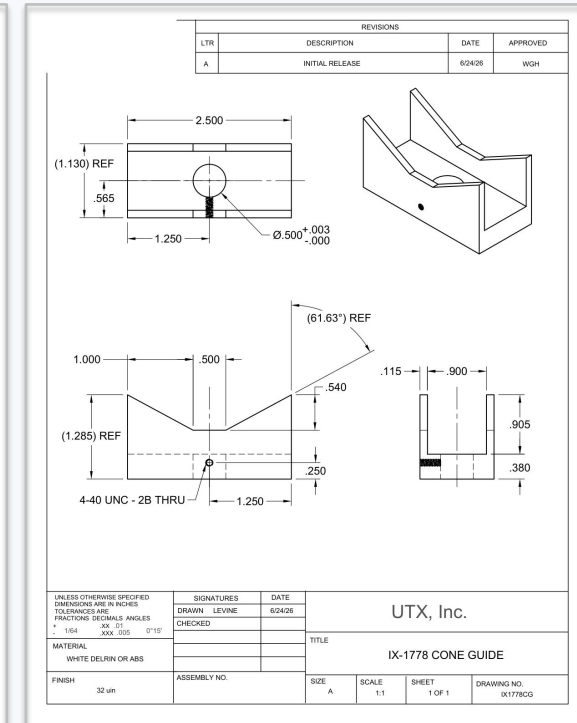
- Converted old Autodesk and hand-drawn machine drawings into 3D CAD models.
- Updated drawings to follow ASME third-angle projection standards.
- Improved consistency and usability for manufacturing.



Transducers manufactured from drawings I created



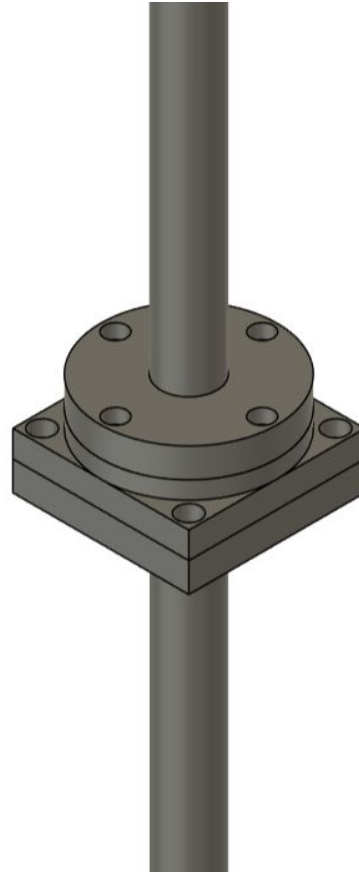
IX-1778 Housing drawing



IX-1778 Cone Guide drawing

Precision Manipulator

- Designed a mechanism to precisely orient an ultrasonic transducer relative to a test target.
- The existing setup could rotate about two axes; I added precise control about a third axis.
- Integrated an optical rotation stage to achieve the required adjustment.



CAD

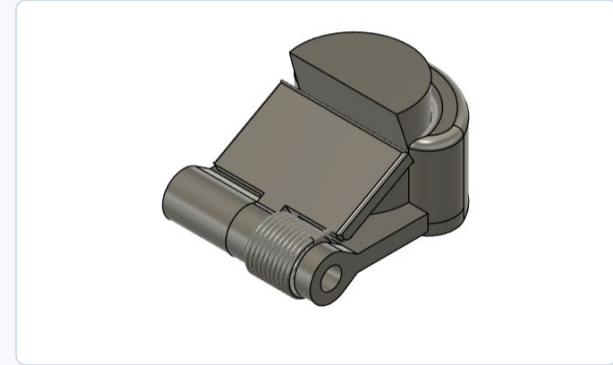
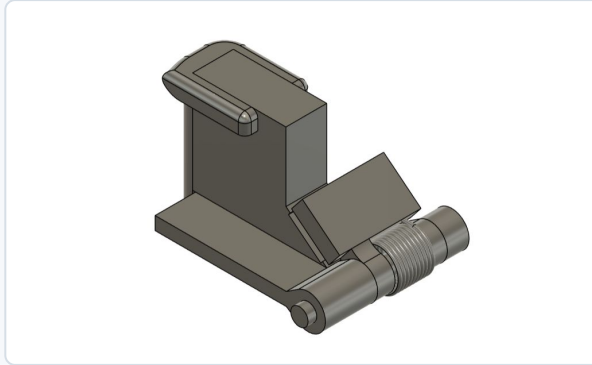


Final assembly

Crystal Clamping Mechanism

- Redesigned the fixture used to attach crystals to wedges.
- Replaced a carefully bent bobby pin setup with a mousetrap-inspired clamping mechanism.
- Designed the mechanism to apply force more evenly across the crystal.

CAD Design

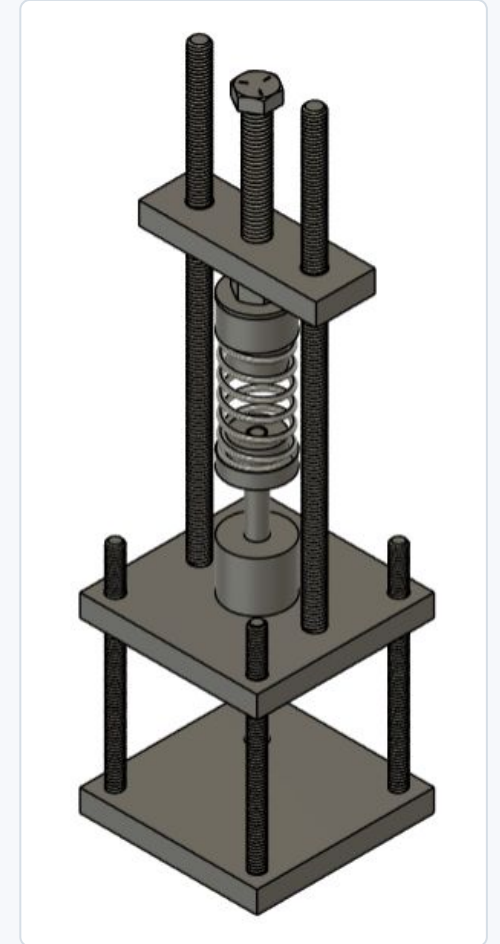
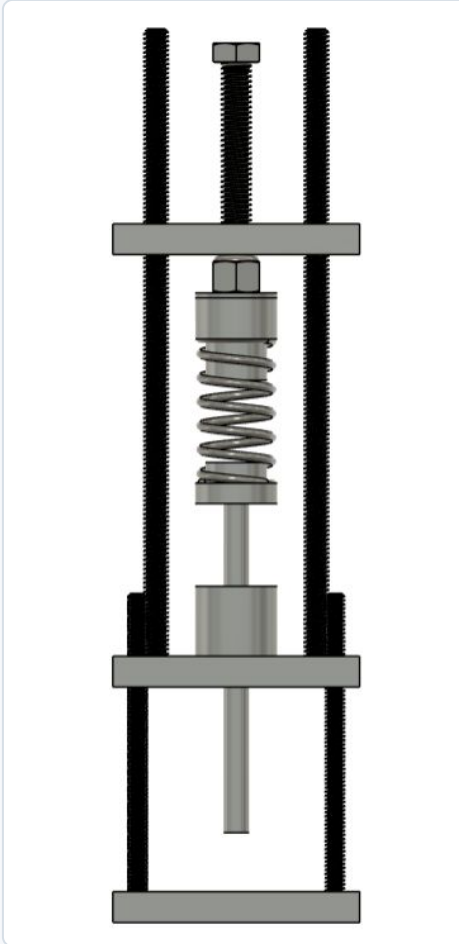


Final Physical Fixture



Spring Press: Reverse Engineering + CAD

- Reverse-engineered a spring press that was likely more than 30 years old.
- Deconstructed the mechanism, improved the design, and created a bill of materials.
- Rebuilt the full assembly in CAD before manufacturing the parts.

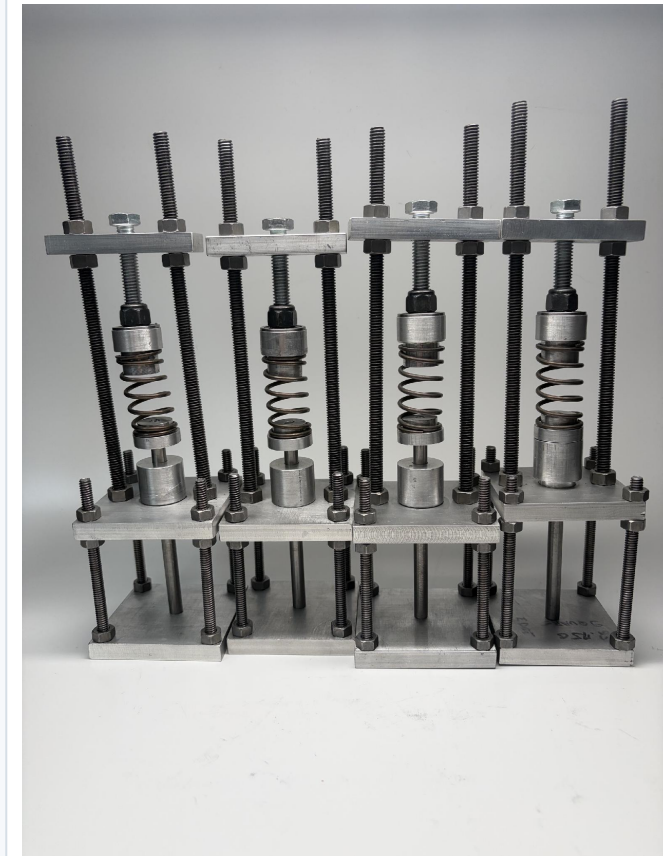


Spring Press: Fabrication

- Manufactured the redesigned press components from the CAD and drawing package.
- Used both a mill and a lathe — two machines I had never used before this internship.
- This became my most time-consuming project and one of the biggest learning experiences of the summer.



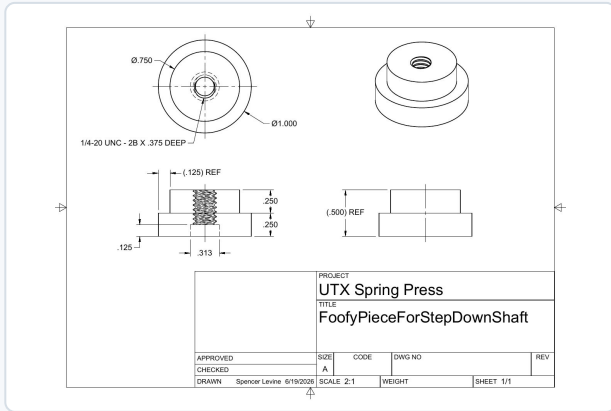
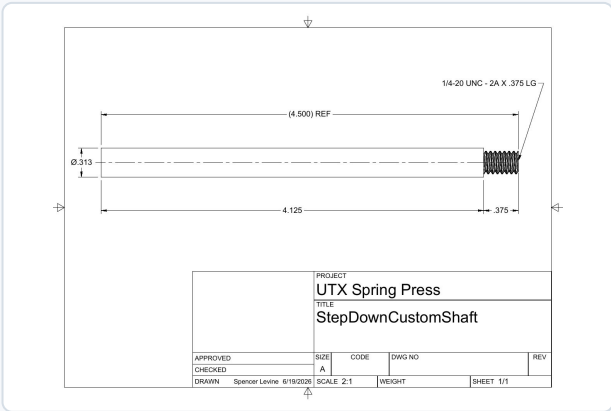
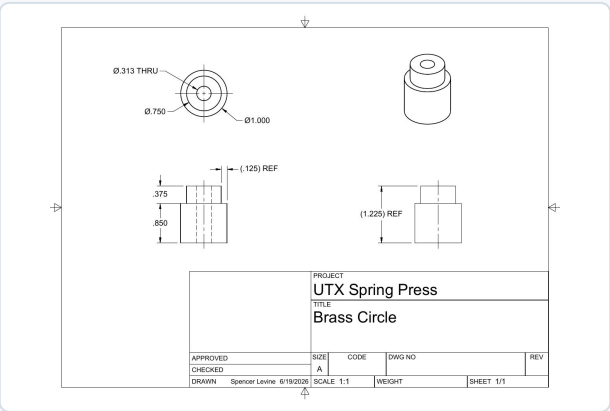
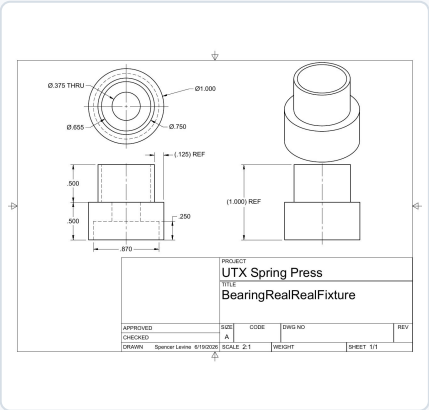
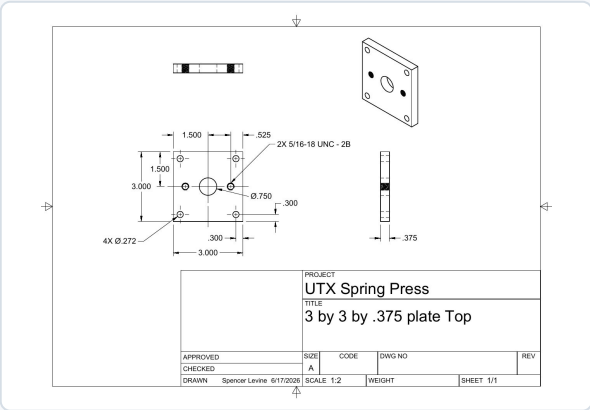
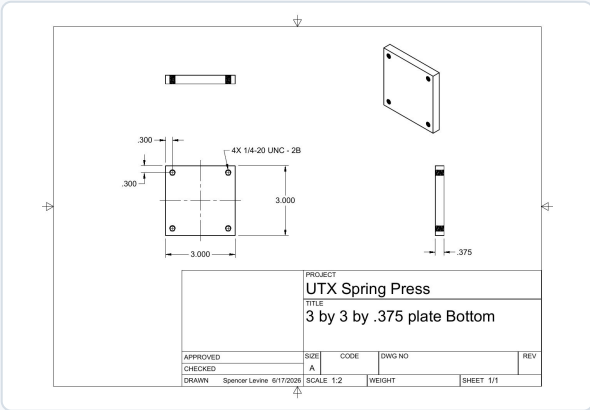
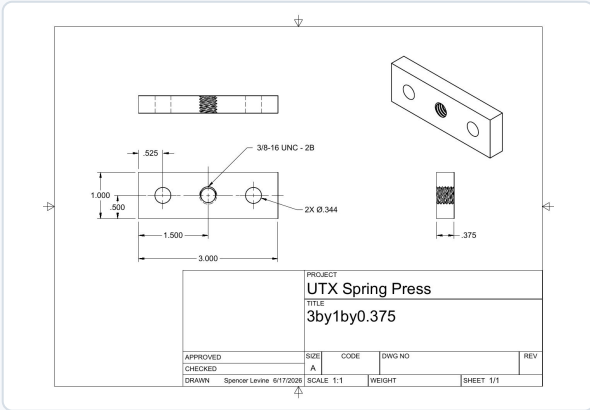
Finished press



Production set

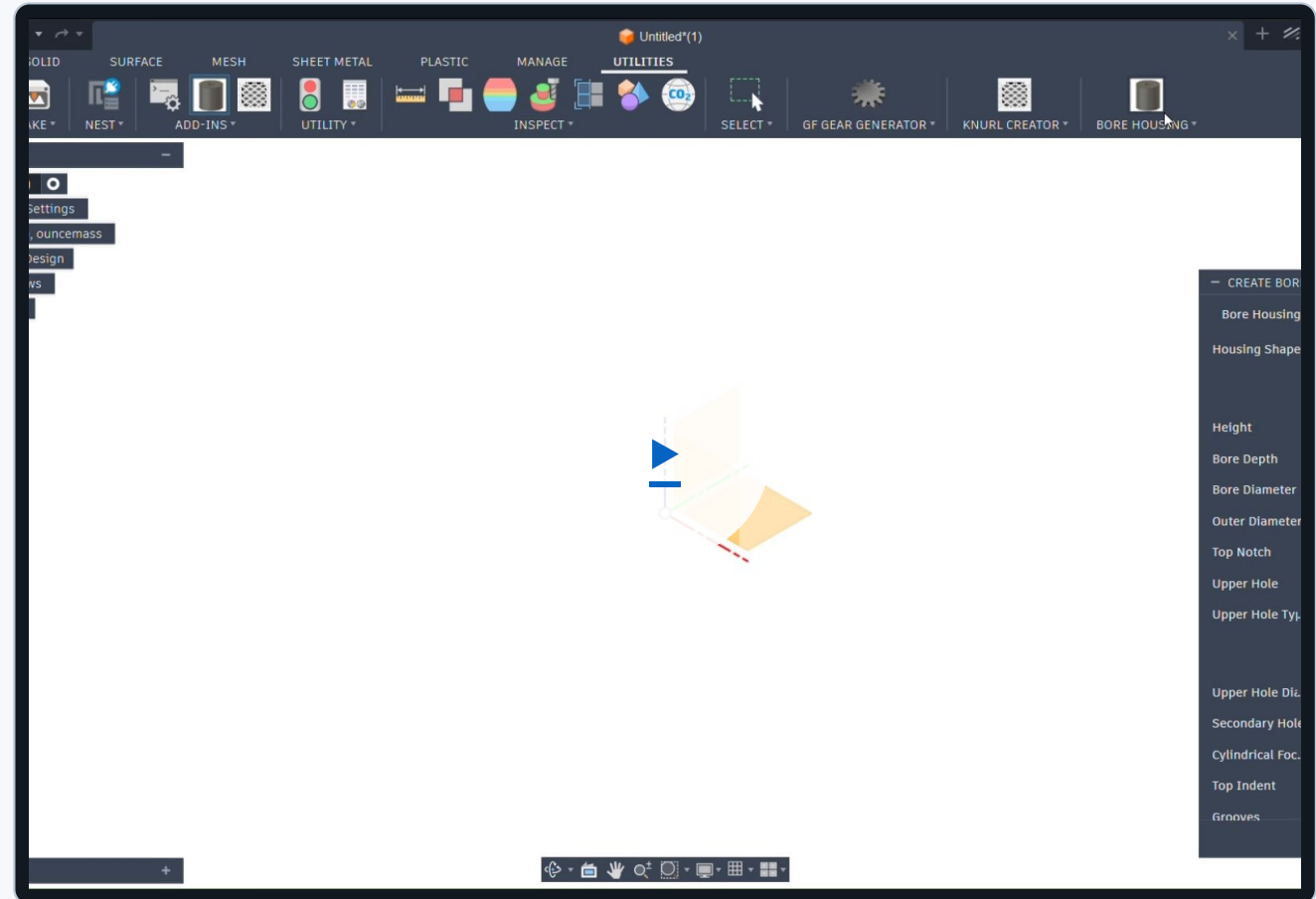
Spring Press: Technical Drawings

Manufacturing-ready drawing package created for the redesigned components.



Fusion 360 Housing Generator

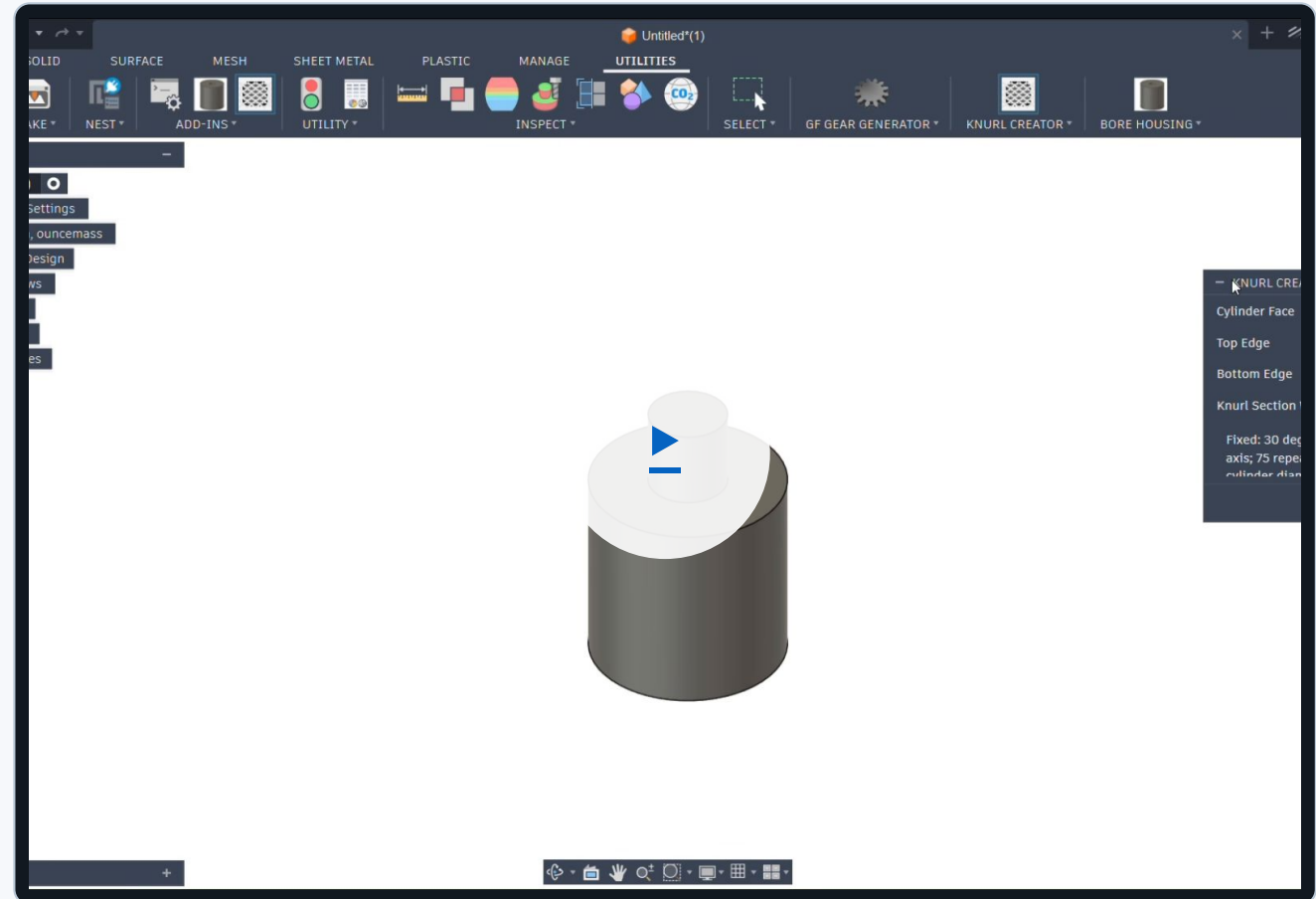
- Noticed that many housing drawings shared the same overall structure and feature set.
- Proposed a custom Fusion 360 plugin to generate the housings parametrically.
- Built the plugin from a parameter list supplied by my boss, using Claude Code as a development tool.



[Click to play video](#)

Fusion 360 Knurl Creator

- Built a second Fusion 360 plugin to automate knurl creation on cylindrical surfaces.
- Turned a repetitive modeling operation into a reusable automated workflow.
- Expanded the automation work beyond the original housing-generator idea.

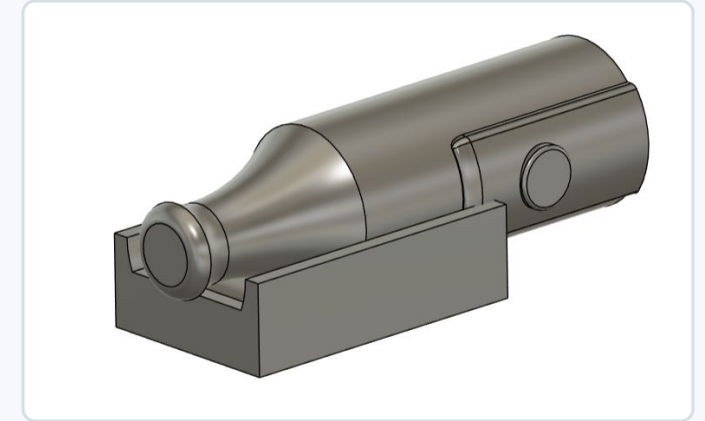
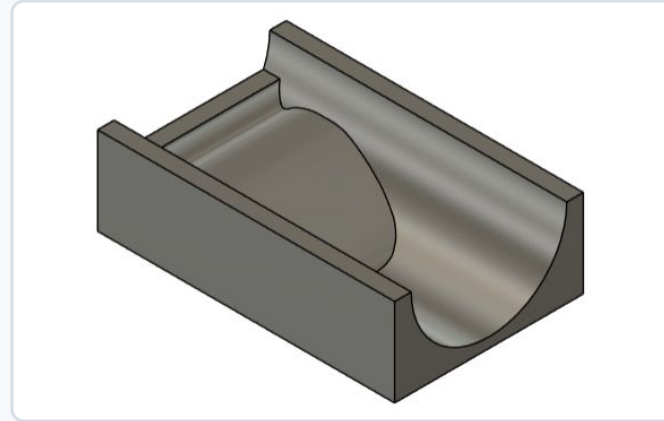


[Click to play video](#)

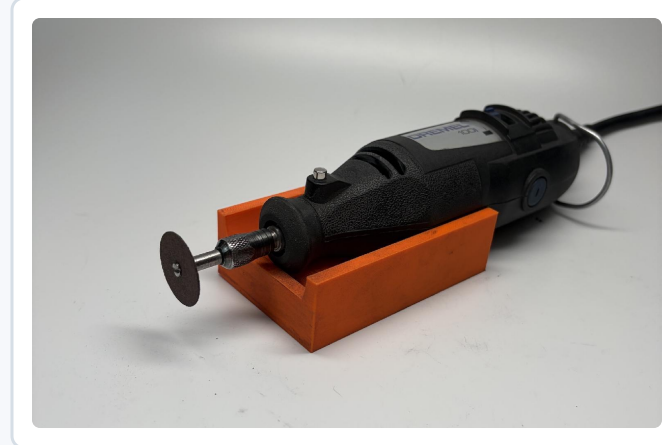
Dremel Cutting Fixture

- Designed a sliding fixture for a Dremel to make straight, repeatable cuts.
- Replaced freehand positioning that could produce angled cuts.
- The fixture improved consistency in a process used to create crystals for the transducers.

CAD Design



Final Physical Fixture



Transducer Manufacturing

- Learned to manufacture transducer subassemblies from start to finish.
- Several steps required magnifying-glass-level precision and careful handling.



Precision assembly work

Transducer Manufacturing

- Used a sandblaster to prepare housing surfaces so epoxy would adhere more effectively.
- Used an engraving machine to add serial numbers and technical information to transducers.



Lathe work

Matching Layer Testing

- Installed quarter-wave matching layers as part of the transducer manufacturing process.
- Verified performance using a pulser-receiver and oscilloscope.
- Captured RF waveforms and used Fast Fourier Transform (FFT) analysis to evaluate performance.



Matching-layer / test work

Teaching Fusion 360

- Trained two UTX employees on Fusion 360.
- One had no prior experience; the other had only used it a few times.
- Focused on practical CAD workflows they could continue using after my internship.



Teaching Fusion

Custom Lens Tooling

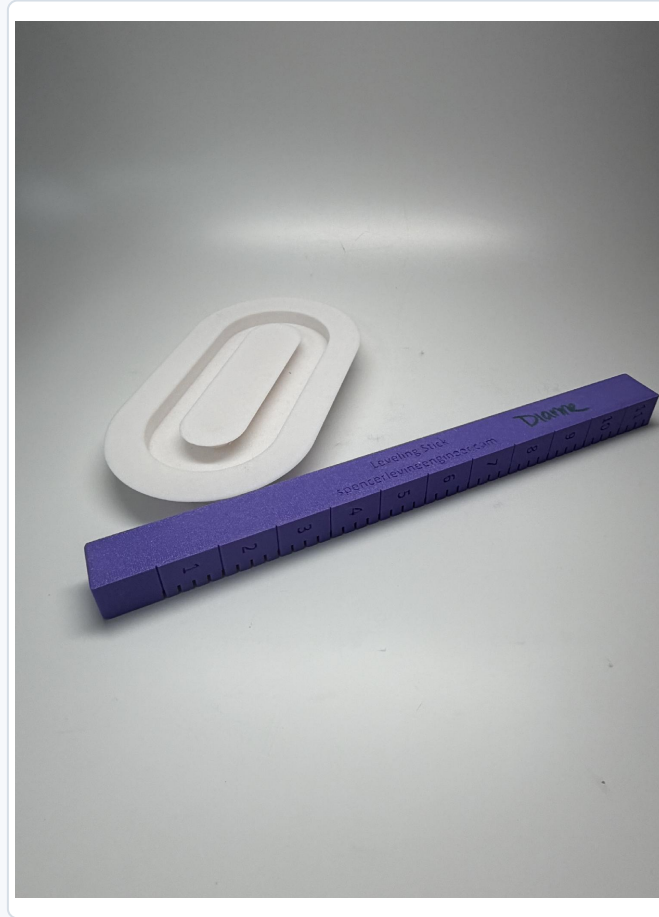
- Designed custom tooling for shaping lenses used on ultrasonic transducers.
- The tooling helped control lens geometry and therefore the way the transducer focused its sound beam.
- Applied fixture-design skills to another production process late in the internship.



Custom lens tooling

Other Design Projects: Pottery + Pool Tools

- Designed custom pottery tools for a studio.
- Designed hooks for baskets and cup holders for an above-ground pool.
- Applied the same CAD and problem-solving process to quick, practical design challenges outside the main transducer work.



Pottery tools



Pool accessories

Invoice + Cost Tracking Database

- Created a database template for company invoices.
- Made it easy to search for an item and view its cost over time.
- Built the tool after brass tubing prices jumped due to tariffs.
- The goal was to make future sudden price increases easier to identify.

Business / operations problem solved with a simple engineering data workflow.

What I Learned

Product Design

Fixtures, mechanisms, and custom tooling

Engineering Drawings

Machine-shop-ready technical documentation

Machining

Hands-on mill and lathe experience

Manufacturing

Precision transducer assembly and production processes

Testing

Oscilloscope, pulser-receiver, RF waveforms, and FFT analysis

Automation

Fusion plugins for repetitive CAD workflows

Communication

Presenting technical ideas to non-technical people, teaching CAD to others

I left UTX with a much better understanding of how design, manufacturing, testing, and communication connect in a real engineering environment.